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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(New Scheme)

(Elect. Engg.)

INSTRUMENTATION TECHNIQUES

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Attempt all questions. Part (a) is compulsory and carries 2 marks. Attempt any* ***two*** *parts from (b), (c) and (d) which carry 7 marks.*

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1. (a) What are the errors involved in measurement process?

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(b) Explain absolute method for testing of potential transformer.

(c) What are the causes of errors in current transformer & explain how these errors can be reduced?

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(d) A 1000/5 A, 50 Hz current transformer has a secondary burden comprising a non-inductive impedance of $1 \cdot 6\ \Omega$. The primary winding has one turn. Calculate the flux in the core & ratio error at full load. Neglect leakage reactance & assume the iron loss in the core to be 1·5 W at full load. The magnetizing MMF is 100 A. (Given, turns ratio is equal to the nominal ratio).

2. (a) Define Transducer. List four physical quantities that a transducer measures.

(b) Describe the construction & working of LVDT with its advantages, disadvantages & applications.

(c) Explain Thermistor in detail.

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(d) A resistance wire strain gauge uses a soft iron wire of small diameter. The gauge factor is +4.2 neglecting

the piezoresistive effects, calculate the Poisson's ratio.

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3. (a) What are recorders & name any two types of graphical recorders?

(b) Describe the generalized data acquisition system with its objectives & applications.

(c) Explain Galvanometer type of recorders, also compare galvanometer type of recorders with potentiometer recorders.

(d) With the help of block-diagram state the different elements of a data logger & explain the function of each element.

4. (a) What is a ladder diagram?

(b) Describe the devices to which PLC, I/P & O/P modules are connected.

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(c) Explain the advantages & disadvantages of PLC.

(d) Explain the procedure for general PLC programming.

5. (a) Explain the register used in PLC in short.

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(b) Describe the relation of digital gate logic to contact/coil logic by a suitable example.

(c) Explain basic PLC functions.

(d) Explain PLC time functions & PLC counter functions.